Response to Reviewer 2

Enhancing Consent Policy Privacy in Blockchain-Based Healthcare Data Sharing

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Working draft. Page and line references will be added after the revised manuscript reaches final pagination.

Comments 1: *The abstract is clear and presents the manuscript; however, it is too long (235 words). It must be less than 200 words.*

Response 1: The abstract has been rewritten and shortened to under 200 words. It also reflects the re-measured padding overhead of the revised system.

Location in the revised manuscript: Abstract.

> In blockchain-based healthcare data sharing, consent policies are expected to be enforced verifiably and auditably, but public-chain transparency may expose sensitive policy information through on-chain representations, verification interfaces, and execution artifacts. Existing privacy-preserving solutions mainly protect data contents, identities, or transactions, not the governing consent policies. We propose the Trusted Proof Service-assisted Zero-Knowledge Policy Interpreter (TPS-ZKPI), which lets an on-chain verifier check authorization decisions while limiting leakage about policy content, identity, and structure. TPS-ZKPI combines fixed-length policy encoding, a reusable zero-knowledge policy interpreter, commitment-based policy binding, and public-chain transcript normalization. We evaluate TPS-ZKPI on Polygon Amoy against metadata-only public-chain adversaries with supervised and unsupervised inference attacks. Under a multi-seed leakage criterion, no attack recovers hidden policy information reproducibly above chance; a paired ablation attributes the gas-channel protection to the gas padding of transcript normalization, at an overhead of about 46% on access and 38% on commit gas.

Comments 2: *Some of the keywords are wrong, such as "policy privacy" – 4 times in the text and "data use ontology" – 7 times.*

Response 2: We understood the comment to mean that the keywords should be the terms the manuscript actually uses, and we revised them accordingly. "policy privacy" is replaced by "consent policy privacy," the phrase of the title, which the Introduction and the Problem section use to name the umbrella objective that the three evaluated objectives (Policy Content Hiding, Policy Indistinguishability, Structure and Parameter Privacy) operationalize; "data use ontology" is written as the standard's name and acronym, "Data Use Ontology (DUO)," as in the Background; "zero-knowledge proof" is given in the plural form used throughout; and "metadata leakage," the manuscript's central concern, is added.

Location in the revised manuscript: Keywords.

> blockchain; consent policy privacy; zero-knowledge proofs; Data Use Ontology (DUO); healthcare data sharing; metadata leakage

Comments 3: *The introduction correctly outlines the issues. However, rows 123–131 point to old sources and methods of encryption. It raises the question "why?" – do the authors not know the newest or do they not want to use the newest; why do the authors not comment on the newest solutions?*

Response 3: The passage (the first paragraph of the Background section in the submitted version) introduced CP-ABE and PRE through their original constructions. We kept those two citations, because the later systems inherit their trust and disclosure properties and that is what the paragraph needs, but we now also cite the current blockchain healthcare designs built on them—PRE-based sharing of medical records and IoMT data (Pei et al., 2024; Zhou et al., 2024) and CP-ABE variants with hidden policies and attributes, distributed key generation, or multi-authority hierarchies (Yang and Liu, 2024; Ding et al., 2026; Wu et al., 2023)—and we state explicitly why the original constructions remain the reference point. The Related Work section discusses the hidden-policy variants and the recent self-sovereign-identity and zero-knowledge authentication work in the same spirit (see also the responses to Reviewer 3, Comment 5, and to Comment 10 below).

Location in the revised manuscript: Section "Background," subsection "Cryptographic Access Control in Blockchain Healthcare," first paragraph.

> Both primitives remain the building blocks of current blockchain healthcare designs: recent schemes combine PRE with on-chain records to delegate access to medical records and IoMT data, and CP-ABE variants with hidden policies and attributes, distributed key generation, or multi-authority hierarchies address the exposure of the access structure itself. We cite the original constructions because the later systems inherit their trust and disclosure properties, which is what matters for this paper: these primitives establish who may obtain the data, and how they relate to *policy* confidentiality on a public chain is discussed in Section 3.

Comments 4: *There is so much empty space on page 6, page 20, and page 25 because of the wrong position of Table 2, Table 9, and Table 12.*

Response 4: The gaps were caused by forcing every float to its source position (`[H]`), which pushes a large table to the next page and leaves the preceding page unfilled. All tables and figures now use the standard `[htbp]` placement, so LaTeX fills each page and floats a large table only when it does not fit. We checked the recompiled manuscript page by page: no page other than the last ends with unused space. Two of the three tables mentioned were also restructured in the revision (the corpus table now lists the revised corpus; the cost table gained a companion scaling table), which further reduced their height.

Location in the revised manuscript: All table and figure environments.

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\begin{table}[htbp]
\begin{figure}[htbp]
```

Comments 5: *The titles of tables and figures are very long. For example, Figure 1 has a 4-row title and Table 12 a 6-row title.*

Response 5: We shortened every caption to one or two sentences that name what is shown, and moved the explanatory material into the body text next to the first reference. For the two

examples: the workflow figure's caption is now "TPS-assisted policy-compliance workflow (steps are described in the text)," with the step numbering explained in the Overview paragraph; the ablation table's caption is now two lines, with the selection rule for its rows and the reading of Δ_{leak} stated in the ablation section. The captions of the corpus, PCH, SPP, PI, clustering, related-work, and cost tables and of the two result figures were shortened in the same way.

Changes in the manuscript. The revised captions are reproduced below; methodological explanations and selection rules now appear in the surrounding prose instead of the captions.

Location in the revised manuscript: Captions of Figures 1, 2, and 3.

> Figure 1. TPS-assisted policy-compliance workflow (steps are described in the text).
>
> Figure 2. Unsupervised K-Means clustering across all three objectives: ARI (top) and NMI (bottom) per target.
>
> Figure 3. Gas-padding ablation over all 45 supervised contrasts: unpadded versus padded deviation from chance, colored by the sign of the 95% interval of Δ_{leak}.

Location in the revised manuscript: Captions of the Privacy Evaluation result and cost tables (Tables 6–16).

> Table 6. Composition of the shared policy corpus (generator seed `unified-decoupled-2026`). DUO terms are multi-label, so their counts overlap; ICD-10 rows cover only the ICD-bearing policies; each SPP axis partitions the full corpus.
>
> Table 7. Headline supervised results on the Polygon Amoy dataset (30 policy-disjoint seeds; a task fails iff $\text{AUC}_{\text{mean}} > 0.55$ and at least 50% of seeds are significant).
>
> Table 8. Policy Content Hiding: per-term supervised attack over 30 policy-disjoint seeds (one-vs-rest tasks; higher-scoring classifier per task). All ten DUO tasks pass.
>
> Table 9. Policy Content Hiding: per-chapter ICD-10 supervised attack over 30 policy-disjoint seeds (higher-scoring classifier per chapter).
>
> Table 10. Policy Indistinguishability: single-execution distinguisher over 30 policy-disjoint seeds, by policy-pair type (higher-scoring classifier per type). `controlled_difference` is reported for reference and excluded from the PI verdict. All gating rows pass.
>
> Table 11. Structure & Parameter Privacy: 21 pairwise contrasts across four annotation axes over 30 policy-disjoint seeds (higher-scoring classifier per contrast).
>
> Table 12. Unsupervised clustering attack (K-Means, K = number of classes, K=2 for pairs) on the on-chain gas features, with the permutation p-value of ARI. PCH clusters all executions globally.
>
> Table 13. Sample-size sensitivity of NMI and ARI under the null hypothesis where all classes are i.i.d. sampled from $\mathcal{N}(0, I_d)$ (T=1000 trials per row).
>
> Table 14. Gas-padding ablation on Polygon Amoy: B1 (no padding) versus B2 (deployed randomized padding) on the same 1119-execution subset; supervised rows are the largest-signal contrasts per experiment (random-forest AUC).
>
> Table 15. Single-point cost profile of the final RC1 configuration at DEPTH = 64. Gas, logs, calldata, and local prover timings are summarized over the 1,773 revised Polygon Amoy executions.
>
> Table 16. Prover-side cost as the policy program length DEPTH grows, with $N = 13$, $M = 11$ and the public interface fixed. On-chain verification cost does not change: the statement keeps five public signals at every depth.

Comments 6: *On row 681 there is a formula in the text. This is not acceptable. Follow the proper format of formulas.*

Response 6: Row 681 of the submitted manuscript belonged to the PCH results paragraph, which reported per-task statistics as inline formulas of the form $\mathrm{AUC}_{\mathrm{mean}} = 0.561 \pm 0.053$. We removed inline formulas from the results prose of the PCH, SPP, and PI sections: statistics are now stated in words with the numeric values ("mean AUC 0.667 with 53% of seeds significant"), and the full figures live in the tables. The remaining displayed formulas are the definitions in the Design and Methodology sections and the paired leakage metric of the ablation, which is now a numbered equation.

Changes in the manuscript. The revised result prose uses short inline numeric quantities rather than sentence-length formulas; the only new formula in this part of the evaluation is the displayed, numbered ablation metric.

Location in the revised manuscript: Section "Privacy Evaluation," subsection "Policy Content Hiding," supervised-attack paragraph.

> Chapter ch5 (diseases of the nervous system) meets both legs: mean AUC 0.667 with 53% of seeds significant. We report it as a FAIL of the criterion and examine below what the classifier separates.

Location in the revised manuscript: Section "Privacy Evaluation," subsection "Policy Indistinguishability," supervised-attack paragraph.

> Pooled over all pairs, the AUC is 0.509 ± 0.035 with 3% of seeds significant. Both gating pair types stay below the 0.55 line: `semantic_equivalent` at 0.549 (7% significant seeds) and `weak_equivalent` at 0.544 (10%), so PI passes.

Location in the revised manuscript: Section "Privacy Evaluation," subsection "Structure & Parameter Privacy," supervised-attack paragraph.

> One contrast, `nestingDepth medium/deep`, meets both legs of the criterion (mean AUC 0.563, 53% significant seeds) and is reported as FAIL. Its separability has the same origin as the PCH ch5 case.

Location in the revised manuscript: Section "Privacy Evaluation," subsection "Ablation: gas-normalization positive control," displayed equation defining Δ_{leak}.

> The paired leakage metric is
>
> $$\Delta_{\mathrm{leak}} = |\mathrm{AUC}_{\mathrm{B1}} - 0.5| - |\mathrm{AUC}_{\mathrm{B2}} - 0.5|,$$
>
> where positive values mean padding moved the attacker closer to chance; we bound each Δ_{leak} with a 95% stability interval over 30 paired train/test splits.

Comments 7: *The opinion of the authors about the results is very short. The conclusions in Part 6 are almost missing.*

Response 7: The Conclusion has been rewritten as four paragraphs: what the revised construction contributes and which soundness gaps it closes; what the evaluation shows and what the

ablation teaches about the un-padded channel; the limitations (trust in the TPS, the supported policy class, the unmodeled channels, the criterion's false-positive rate); and three directions that follow from those limitations. The Discussion likewise opens with the authors' reading of the results, including the two criterion failures and why we report rather than absorb them, before turning to the trust model.

Changes in the manuscript. The revised Discussion interprets the empirical findings before turning to architectural trade-offs, and the Conclusion now covers contribution, findings, limitations, and future directions in four distinct paragraphs.

Location in the revised manuscript: Section "Discussion," opening paragraph.

> The evaluation supports a scoped claim: against the modeled public-chain metadata adversary, TPS-ZKPI provides empirical resistance rather than proven policy confidentiality. Of the 45 supervised tasks, 43 stay below the multi-seed leakage criterion, and unsupervised clustering shows no recoverable structure beyond one small, permutation-significant nesting-depth ARI. The two tasks that meet the criterion—one ICD-10 chapter and the `nestingDepth` `medium/deep` contrast—are reported as failures, and both are carried by the random pad target rather than by the policy: subtracting the logged target from the access gas returns each to PASS, and replacing the gas features by policy-independent draws reproduces such failures in four of ten replicates. This is a property of the criterion on a corpus in which coverage-layer policies are executed once, so the thirty seeds resample the same random draws; we keep the criterion as stated and document the diagnostic, since lowering the significance threshold or raising the AUC threshold to absorb these cases would weaken it for genuine leaks. The gas-padding ablation attributes the gas-channel protection to randomized gas padding: without it the same attacks recover the policy family with AUCs up to 0.90. Content hiding itself rests on the commitment-only interface and the content-independent transcript normalization; padding is the defense-in-depth layer that absorbs whatever deterministic gas differences remain, of which the ablation exhibits one that does not come from the policy logic at all but from the byte encoding of a public transaction argument. Together with fixed-length policy representation, reusable interpretation, and commitment binding, these mechanisms limit the extent to which public-chain observers infer hidden policy content or structural properties from authorization transactions.

Location in the revised manuscript: Section "Conclusion" (entire section).

> This paper studied policy-private consent enforcement in blockchain-based healthcare data sharing under public-chain observability and an explicit TPS trust assumption. We proposed TPS-ZKPI, a Trusted Proof Service-assisted zero-knowledge approach that binds hidden provider-defined policies to on-chain commitments, evaluates Boolean-based consent policies through a reusable policy interpreter, and normalizes public verification artifacts to reduce metadata-level leakage. The revised construction closes two soundness gaps of the original prototype: the program length is part of the committed and proved relation, so an authorizing proof cannot rest on a shorter valid prefix, and the proof is bound to the requester pseudonym, resource, and session nonce, so a captured proof cannot be replayed under fresh parameters within a deployment. Profile-backed data-use terms are derived in-circuit from an issuer-signed credential, which separates what the proof establishes cryptographically from what remains a matter of issuer governance.

> The evaluation shows what such a design can and cannot promise. On a live testnet corpus, a supervised attacker with policy-disjoint training data and an unsupervised clustering attacker recover neither policy content, nor policy identity, nor structure and parameters from the normalized transcript beyond what the padding's own random draws produce, and the paired ablation shows that the padding is what removes the last policy-correlated gas signal. The ablation also delivers a cautionary result: the un-padded channel it exposes is not the policy logic, which the fixed circuit already hides, but the byte-level pricing of one public transaction argument—a reminder that on a public chain any deterministic encoding difference that correlates with the policy is a channel, and that the defense must cover the class rather than the instances one anticipated.
>
> The results hold for a metadata-only public-chain adversary and are empirical. Policy confidentiality against the TPS itself is not cryptographic but rests on the service following its workflow, and the evaluation's provider-operated TPS concentrates that trust at the provider. The supported policy class is an access-time checker for bounded Boolean policies over standardized terms: institutional or ethics approval, downstream purpose limitation, post-access duties, delegation, and credential revocation stay outside the proved statement. On-chain policy revocation and evolution govern future authorization decisions but do not retract data already released or enforce downstream cessation or deletion. Timing, ordering, mempool, and off-chain channels, adaptive probing, and correlations between outcomes and policy classes are not modeled, and the leakage criterion has a measurable false-positive rate on corpora that execute each policy once.
>
> Three directions follow. Reduced-trust proof generation—collaborative SNARKs or attested execution—would relax the trust concentration that the TPS introduces and, with a holder-signed authorization context, remove the service's ability to reuse valid evidence. A governed issuer registry with in-circuit membership and revocation witnesses would turn the credential path into an institutional guarantee. And extending the evaluation to active adversaries and to the timing and ordering channels excluded here would test whether transcript normalization remains sufficient once the observer is no longer passive.

Comments 8: *The English language looks quite good. The title and abstract reflect the content.*

Response 8: We thank the reviewer. The title is unchanged; the abstract was shortened and updated as described under Comment 1.

Comments 9: *The discussion and conclusion summarize the proposed idea. However, there is a repetition of the same conclusion in the abstract, discussion, and conclusion. This repetition of the same information should be avoided.*

Response 9: The three texts now serve different purposes. The abstract states the result and the overhead figures once. The Discussion interprets the results—the scope of the claim, the two criterion failures and their origin, the cost of padding, the trust model, and the attribute-origin guarantee—without restating the abstract. The Conclusion no longer repeats the numbers; it summarizes the contribution, the meaning of the results (including the cautionary finding of the ablation), the limitations, and the directions that follow.

Changes in the manuscript. Representative passages showing the distinct role of each section are reproduced below.

Location in the revised manuscript: Abstract.

> In blockchain-based healthcare data sharing, consent policies are expected to be enforced verifiably and auditably, but public-chain transparency may expose sensitive policy information through on-chain representations, verification interfaces, and execution artifacts. Existing privacy-preserving solutions mainly protect data contents, identities, or transactions, not the governing consent policies. We propose the Trusted Proof Service-assisted Zero-Knowledge Policy Interpreter (TPS-ZKPI), which lets an on-chain verifier check authorization decisions while limiting leakage about policy content, identity, and structure. TPS-ZKPI combines fixed-length policy encoding, a reusable zero-knowledge policy interpreter, commitment-based policy binding, and public-chain transcript normalization. We evaluate TPS-ZKPI on Polygon Amoy against metadata-only public-chain adversaries with supervised and unsupervised inference attacks. Under a multi-seed leakage criterion, no attack recovers hidden policy information reproducibly above chance; a paired ablation attributes the gas-channel protection to the gas padding of transcript normalization, at an overhead of about 46% on access and 38% on commit gas.

Location in the revised manuscript: Section "Discussion," opening and scope paragraphs.

> The evaluation supports a scoped claim: against the modeled public-chain metadata adversary, TPS-ZKPI provides empirical resistance rather than proven policy confidentiality. Of the 45 supervised tasks, 43 stay below the multi-seed leakage criterion, and unsupervised clustering shows no recoverable structure beyond one small, permutation-significant nesting-depth ARI. The two tasks that meet the criterion—one ICD-10 chapter and the `nestingDepth medium/deep` contrast—are reported as failures, and both are carried by the random pad target rather than by the policy: subtracting the logged target from the access gas returns each to PASS, and replacing the gas features by policy-independent draws reproduces such failures in four of ten replicates. The scope of this claim is the adversary of Section 6.1: a passive observer of commitments, proofs, public parameters, calldata, events, and gas, on a corpus of generated policies. Timing and ordering of transactions, mempool and broadcast behaviour, off-chain communication with the TPS, adaptive requester probing, repeated access attempts, correlations between authorization outcomes and policy classes (every request in the corpus is granted), and compromised TPS behaviour are not modeled, and the results should not be read as establishing general policy confidentiality against them.

Location in the revised manuscript: Section "Conclusion," contribution and future-directions paragraphs.

> This paper studied policy-private consent enforcement in blockchain-based healthcare data sharing under public-chain observability and an explicit TPS trust assumption. We proposed TPS-ZKPI, a Trusted Proof Service-assisted zero-knowledge approach that binds hidden provider-defined policies to on-chain commitments, evaluates Boolean-based consent policies through a reusable policy interpreter, and normalizes public verification artifacts to reduce metadata-level leakage. The revised construction closes two soundness gaps of the original prototype: the program length is part of the committed and proved relation, so an authorizing proof cannot rest on a shorter valid prefix, and the proof is bound to the requester pseudonym, resource,

> and session nonce, so a captured proof cannot be replayed under fresh parameters within a deployment. Profile-backed data-use terms are derived in-circuit from an issuer-signed credential, which separates what the proof establishes cryptographically from what remains a matter of issuer governance.
>
> Three directions follow. Reduced-trust proof generation—collaborative SNARKs or attested execution—would relax the trust concentration that the TPS introduces and, with a holder-signed authorization context, remove the service's ability to reuse valid evidence. A governed issuer registry with in-circuit membership and revocation witnesses would turn the credential path into an institutional guarantee. And extending the evaluation to active adversaries and to the timing and ordering channels excluded here would test whether transcript normalization remains sufficient once the observer is no longer passive.

Comments 10: *Many of the references are old [7, 8, 10, 14, 15, 16, 17, 18, 19, 21, 23, 24, 25, 26, 27, 30, 31]. Some of them do not follow the proper format. I offer the authors to find newer sources and replace these old sources.*

Response 10: We reviewed each of the cited entries and took two actions.

Newer sources. Seven references from 2024–2026 were added and are now cited alongside the older ones wherever the text relies on them: PRE-based blockchain sharing of medical records and IoMT data (Pei et al., *Computer Networks*, 2024; Zhou et al., *Peer-to-Peer Networking and Applications*, 2024), hidden-policy and policy-hiding CP-ABE for medical data (Yang and Liu, *IET Information Security*, 2024; Ding et al., *IEEE J. Biomed. Health Inform.*, 2026), holder-controlled access to claims in self-sovereign identity (Buccafurri et al., *Blockchain: Research and Applications*, 2024), zero-knowledge anonymous authentication of patients (Madine et al., *Array*, 2025), and a survey of zero-knowledge proofs for blockchain privacy (Shashidhara et al., *Security and Privacy*, 2024).

Retained older sources. The remaining entries in the reviewer's list are the original constructions or the standards the manuscript builds on, for which no newer source is the appropriate citation: Groth16 [18] is the proof system deployed; Zerocash, Zeth, and AZTEC [7, 8, 19] are the transaction-privacy systems the Introduction contrasts with; CP-ABE, PRE, and functional encryption [10, 16, 17, 21] are the primitives whose disclosure properties the Background discusses (now paired with the 2024–2026 systems above); Plonk and LegoSNARK [26, 27] are the reusable proving systems of the Related Work; Ekiden [23], the XACML privacy work [24], and multi-party policy evaluation [25] are the representative designs of their respective lines; and ICD-10 [14], the GA4GH consent codes [15], OBI [30], and the Jaccard index [31] are the standards and definitions the evaluation uses. We kept them for that reason and cite them for that reason only.

Format. The AZTEC entry is now a properly formatted technical whitepaper with its URL and access date, the Zeth entry an arXiv e-print with identifier, and the Ibraimi et al. entry a University of Twente technical report (the submitted version carried a wrong journal name); all new entries include DOIs.

Changes in the manuscript. The revised passages and the seven added references are reproduced below.

Location in the revised manuscript: Section "Background," subsection "Cryptographic Access Control in Blockchain Healthcare," first paragraph.

> Fine-grained consent management in blockchain-based healthcare data sharing is

> commonly built on cryptographic access-control primitives, chiefly Ciphertext-Policy Attribute-Based Encryption (CP-ABE) and Proxy Re-Encryption (PRE). In CP-ABE, the encrypting party embeds an attribute-based access policy into the ciphertext, so that only requesters whose credentials satisfy the policy can decrypt the protected data. PRE lets a semi-trusted intermediary transform a ciphertext encrypted for one party into one decryptable by a designated requester, delegating access without exposing the plaintext. Both primitives remain the building blocks of current blockchain healthcare designs: recent schemes combine PRE with on-chain records to delegate access to medical records and IoMT data, and CP-ABE variants with hidden policies and attributes, distributed key generation, or multi-authority hierarchies address the exposure of the access structure itself. We cite the original constructions because the later systems inherit their trust and disclosure properties, which is what matters for this paper: these primitives establish who may obtain the data, and how they relate to *policy* confidentiality on a public chain is discussed in Section 3.

Location in the revised manuscript: Section "Related Work," paragraph on hidden-policy access control.

> Hidden-policy access-control work reduces direct policy exposure by using encrypted policies, hidden-policy ABE variants, off-chain matching, or policy-holder components. Functional encryption gives a more general cryptographic abstraction for revealing function outputs while hiding inputs, but it does not by itself yield publicly verifiable authorization evidence. Private-state blockchain systems, including Secret Network and Oasis/Ekiden, protect smart-contract state or execution through encrypted state and trusted execution environments. These lines of work address adjacent forms of policy or state hiding, but TPS-ZKPI targets a more specific setting: hiding consent policies on a public chain while retaining public verifiability of each authorization decision. Even where policies are hidden, existing designs may still leak policy-dependent artifacts through ciphertext structures, policy-update transactions, authorization workflows, or repeated interactions. Moreover, many target permissioned or off-chain settings, where enforcement artifacts are not globally visible as they are on public blockchains.

Location in the revised manuscript: Section "Related Work," new paragraph on self-sovereign-identity architectures.

> Self-sovereign-identity (SSI) architectures move authorization to verifiable claims held by the requester and shift the question to whom the holder may trust with those claims. Buccafurri et al. combine smart contracts with ciphertext-policy attribute-based proxy re-encryption so that a holder's claims can be read only by verifiers that satisfy a holder-defined policy, with the contract keeping verifiers accountable for the access they obtain; zero-knowledge presentations of credentials likewise let a patient authenticate anonymously against public or private chains, and surveys of zero-knowledge proofs for blockchain privacy catalogue this design space. TPS-ZKPI sits on the other side of the same authorization relation. In SSI designs the protected object is the requester's claim set and the policy that governs it is the holder's, whereas here the protected object is the provider's consent policy: the requester's claims are consumed as private witnesses of a proof whose statement is fixed and whose verifier is a public contract, so verifier authorization is not a design

question (any observer may verify) and accountability is carried by the recorded, context-bound outcome rather than by a contract that meters the release of claims. The trust the two settings place off-chain also differs: an SSI verifier that decrypts a claim learns it, whereas the TPS learns both policy and claims, which is why Section 4 treats the TPS as an explicit trust boundary and Table 3 states what survives its compromise. Combining the two—holder-authorized presentation of the claims that TPS-ZKPI consumes as witnesses, including a per-request signature over the authorization context—would prevent a TPS that observes one presentation from reusing the evidence for fresh sessions, as discussed in Section 7.

Location in the revised manuscript: Reference list, seven new entries from 2024–2026.

Pei, H.; Yang, P.; Li, W.; Du, M.; Hu, Z. Proxy Re-Encryption for Secure Data Sharing with Blockchain in Internet of Medical Things. *Computer Networks* 2024, 245, 110373. https://doi.org/10.1016/j.comnet.2024.110373.

Zhou, H.; Si, X.-M.; Zhao, B.; Zhang, Y.; Qu, Y. Secure electronic medical records sharing scheme based on blockchain by using proxy re-encryption. *Peer-to-Peer Networking and Applications* 2024, 17, 4197–4209. https://doi.org/10.1007/s12083-024-01775-y.

Yang, X.; Liu, J. HA-Med: A Blockchain-Based Solution for Sharing Medical Data with Hidden Policies and Attributes. *IET Information Security* 2024, 2024, 2498245. https://doi.org/10.1049/2024/2498245.

Ding, X.; Liu, Y.; Ning, J.; Chen, D. Blockchain-Enhanced Anonymous Data Sharing Scheme for 6G-Enabled Smart Healthcare With Distributed Key Generation and Policy Hiding. *IEEE Journal of Biomedical and Health Informatics* 2026, 30, 666–675. https://doi.org/10.1109/JBHI.2025.3550261.

Buccafurri, F.; De Angelis, V.; Nardone, R. How can the holder trust the verifier? A CP-ABPRE-based solution to control the access to claims in a Self-Sovereign-Identity scenario. *Blockchain: Research and Applications* 2024, 5, 100196. https://doi.org/10.1016/j.bcra.2024.100196.

Madine, M.; Salah, K.; Jayaraman, R.; Yaqoob, I. Zero-knowledge proofs for anonymous authentication of patients on public and private blockchains. *Array* 2025, 28, 100590. https://doi.org/10.1016/j.array.2025.100590.

Shashidhara, R.; Chirakarotu Nair, R.; Kumar Panakalapati, P. Promise of Zero-Knowledge Proofs (ZKPs) for Blockchain Privacy and Security: Opportunities, Challenges, and Future Directions. *Security and Privacy* 2024, 8, e461. https://doi.org/10.1002/spy2.461.